Assignment: AI Agent Formulation and Analysis

RAILWISE: RAILWAY PLATFORM ALLOCATION AI AGENT

1. Agent Overview

RailWise is an intelligent railway platform-management system that assists station staff in assigning appropriate platforms to arriving and departing trains. It evaluates factors such as train schedules, platform availability, train-specific requirements, delays, and temporary closures to suggest feasible platform assignments. Its goals are to minimize platform conflicts, reduce avoidable waiting times, and help.

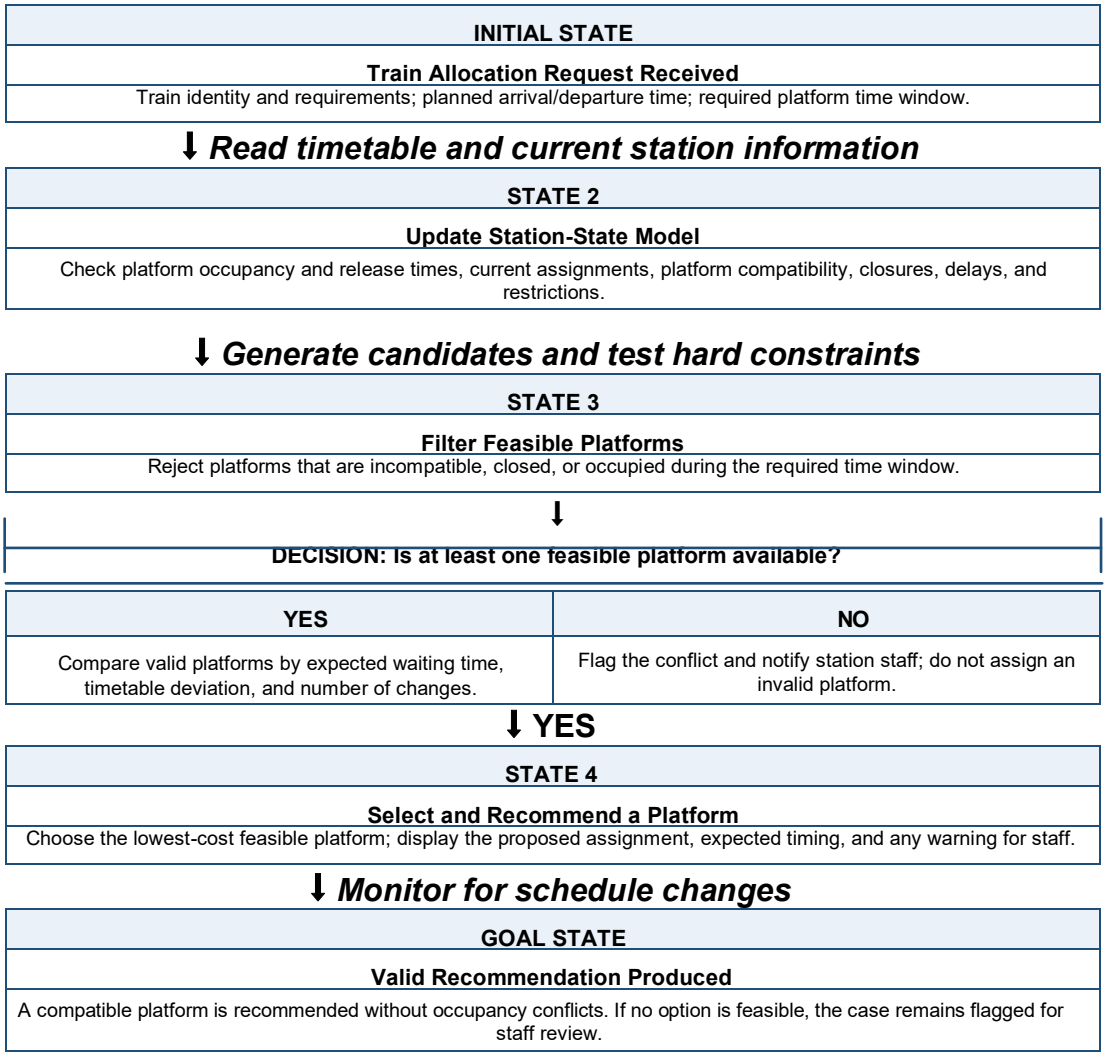
When a delay or closure occurs, RailWise reviews affected assignments and checks alternatives. If no valid option exists, it flags the case for staff review.

2. Agent Type

RailWise is a model-based, utility-based agent. It maintains an updated view of trains, platforms, and current assignments, then compares feasible choices. It prefers options with lower waiting time and fewer schedule changes, but never ignores hard constraints such as platform compatibility or overlapping occupancy.

3. Graphical Representation

The flowchart represents the allocation process as states connected by actions, with a feasibility decision and a staff-review path when no valid platform is available.



Feedback: New arrivals, departures, delays, or closures update the station-state model and restart the checks. RailWise supports staff decisions; it does not control signals or authorize train movements.

4. Problem Formulation

Railway platform allocation is a sequential scheduling and constraint-satisfaction problem. Each assignment must fit the train's requirements and the platform's available time window, while accounting for decisions already made for other trains.

Problem type	Sequential scheduling and constraint-satisfaction problem.
States	Train arrival/departure times and status, platform occupancy, assignments, delays, and restrictions.
Initial state	Published timetable, platform details, current occupancy, train requirements, and known disruptions.
Actions	Check compatibility, assign or reassign a train, update station information, and notify staff about conflicts.
Transition model	Arrivals, departures, delays, assignments, and closures update the station state and may require allocations to be checked again.
Goal test	Each train has a compatible platform for its required time, without overlapping use or violated restrictions; infeasible cases are flagged for staff.
Path cost	Minimize waiting time, timetable deviation, and unnecessary platform changes while preserving hard constraints.

5. Environment Properties

• **Partially observable:** Updates may not arrive immediately, so the agent's view can be incomplete.

• **Stochastic:** Unexpected delays, equipment issues, or operational changes can affect the schedule.

• **Sequential:** One train's platform assignment can limit choices for later trains.

• **Dynamic:** Train timings, occupancy, and restrictions can change while the agent operates.

• **Hybrid:** Platform choices are discrete, while times and delays are numerical.
• **Multi-agent:** Drivers, station staff, and railway control systems also influence the environment.

6. PEAS Framework

Performance measure	Avoid platform conflicts, respect restrictions, reduce waiting and timetable deviation, limit unnecessary reallocations, and flag cases requiring human review.
Environment	Railway station, trains, platforms, timetables, station staff, and changing operating conditions.
Actuators	Proposed platform assignments, staff alerts, schedule display, and allocation reports.
Sensors	Timetable, arrival/departure updates, platform occupancy, train requirements, closures, and delay reports.

7. Example Scenario

Suppose a train is due while Platform 1 is still occupied. RailWise rules out Platform 1, checks other platforms for availability and compatibility, and recommends Platform 2 if it is free and suitable. If no platform meets the requirements, it alerts staff for manual review.